POE Aerospace Rover Project Brief Packet

Unit 0-5 student-facing project briefs for the Principles of Engineering rover systems pathway.

Aerospace Readiness Mini-Challenge

Unit 0 Project Brief

Engineering Scenario

Before teams begin the major rover systems work, each student must demonstrate readiness with safety, documentation, CAD refreshers, and basic systems thinking.

Design Goal

Complete certifications and produce a small support artifact that proves you can document, fabricate, test, and reflect like an engineering team member.

Criteria	• Evidence is student-facing and organized in the engineering notebook or packet. • Prototype or support artifact connects to a future rover or mission-support need. • Documentation includes problem, criteria, constraints, sketch, build evidence, and reflection.
Constraints	• Use only approved classroom tools and materials. • Follow FabLab safety rules and teacher approval requirements. • The artifact must be small enough to finish within the Unit 0 timeline.
Required Evidence	• Certification readiness notes • Mini-challenge sketch or CAD evidence • Build/test photo or observation notes • Short reflection on readiness for POE projects
Checkpoints	• Identify readiness expectations • Complete tool/safety evidence • Build or document the mini-challenge artifact • Submit final reflection

Final Design Review Expectations

Your final review should clearly connect the problem, evidence, prototype decisions, testing results, limitations, and next recommended improvement. Strong teams show how data changed the design rather than only describing what they built.

Student Deliverable Reminder

Keep all sketches, calculations, data tables, photos, revision notes, and decisions in your engineering notebook or assigned project packet. The notebook should make your design process visible.

VEX Rover Mechanisms Challenge

Unit 1 Project Brief

Engineering Scenario

Planetary rovers need mechanisms that can lift, deploy, steer, transfer, clamp, or position parts reliably while using limited power and mass.

Design Goal

Design, build, and test a VEX mechanism that redirects force, speed, distance, or motion for a rover mission-support task.

Criteria	• Mechanism performs a clear rover-related task. • Team explains mechanical advantage, energy transfer, or power trade-offs. • Prototype performance is measured with repeated tests. • Final recommendation is based on data and observed limitations.
Constraints	• Use VEX parts and approved fasteners only unless the teacher approves an add-on. • Prototype must be safe to operate on the tabletop or floor test area. • Testing must include at least three repeated trials.
Required Evidence	• Problem statement and mechanism concept sketches • System diagram showing input, process, and output • Mechanical calculation or trade-off explanation • Test data and revision notes • Final design review summary
Checkpoints	• Define the mechanism task • Generate team concepts • Select and build one concept • Test and revise • Present the engineering recommendation

Final Design Review Expectations

Your final review should clearly connect the problem, evidence, prototype decisions, testing results, limitations, and next recommended improvement. Strong teams show how data changed the design rather than only describing what they built.

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Rover Payload Support Structure Challenge

Unit 2 Project Brief

Engineering Scenario

A rover must carry payloads, sensors, batteries, and mission tools while staying light enough to move efficiently and strong enough to survive testing.

Design Goal

Design, fabricate, and test a lightweight support structure that balances strength, stiffness, mass, material choice, and manufacturability.

Criteria	• Structure supports the required payload without unsafe failure. • Design shows thoughtful use of geometry, material properties, and load paths. • Testing includes measured mass, load, deflection, and observations. • Final claim explains strength-to-weight performance.
Constraints	• Use approved classroom materials and fabrication tools. • Structure must fit within the teacher-provided size envelope. • Teams must document setup and safety before load testing.
Required Evidence	• Annotated sketches or CAD concept • Material and geometry rationale • Load-test setup diagram • Data table with repeated measurements • Failure mode or limitation analysis
Checkpoints	• Analyze loads and constraints • Choose a structural concept • Build and inspect prototype • Run load/deflection tests • Defend the structure using evidence

Final Design Review Expectations

Your final review should clearly connect the problem, evidence, prototype decisions, testing results, limitations, and next recommended improvement. Strong teams show how data changed the design rather than only describing what they built.

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Autonomous Rover Mission + Ground Support System

Unit 3 Project Brief

Engineering Scenario

Autonomous rover missions rely on control logic, sensors, feedback, actuators, and support systems to complete a repeatable task safely.

Design Goal

Program or plan an autonomous rover mission and connect it to a VEX, fluid-power, or physical ground-support system.

Criteria	• Mission includes clear start, route/task, and end conditions. • Team explains open-loop and/or closed-loop control behavior. • Ground-support system connects meaningfully to the rover mission. • Testing evidence shows reliability, debugging, and iteration.
Constraints	• Use approved VEX/robotics equipment and classroom test zones. • Follow battery, reset, and traffic-flow expectations during testing. • No team may test without a safe mission path and observer role.
Required Evidence	• Mission flowchart or pseudocode • Input/output table or sensor/actuator diagram • Prototype or code evidence • Test log with revisions • Final mission review with performance data
Checkpoints	• Plan the mission logic • Map sensors, inputs, and outputs • Build or code the rover task • Test and debug one variable at a time • Demonstrate and defend the system

Final Design Review Expectations

Your final review should clearly connect the problem, evidence, prototype decisions, testing results, limitations, and next recommended improvement. Strong teams show how data changed the design rather than only describing what they built.

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Rover Performance Data Investigation

Unit 4 Project Brief

Engineering Scenario

Engineers use repeated trials and statistics to decide whether a rover system is reliable, predictable, and ready for more advanced testing.

Design Goal

Collect and analyze rover or mission-performance data using repeated trials, graphs, descriptive statistics, and kinematics when appropriate.

Criteria	• Investigation defines independent, dependent, and controlled variables. • Data set includes enough repeated trials to discuss variation. • Graphs and statistics support a clear engineering claim. • Conclusion acknowledges uncertainty, error, and next test steps.
Constraints	• Use an approved test setup and measurement method. • Keep test conditions consistent unless changing the independent variable. • Show calculations clearly enough for another team to check them.
Required Evidence	• Investigation question and variables • Procedure and safety notes • Data table and graph • Statistics and sample calculation • Claim-evidence-reasoning conclusion
Checkpoints	• Choose the test question • Plan variables and data collection • Run repeated trials • Analyze results • Write the engineering conclusion

Final Design Review Expectations

Your final review should clearly connect the problem, evidence, prototype decisions, testing results, limitations, and next recommended improvement. Strong teams show how data changed the design rather than only describing what they built.

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Integrated Aerospace Rover Systems Capstone

Unit 5 Project Brief

Engineering Scenario

A complete rover system combines mechanisms, structures, controls, fabrication, testing, and documentation into one mission-ready engineering solution.

Design Goal

Design, build, test, revise, and defend an integrated rover system that solves a defined aerospace mission problem.

Criteria	• System includes at least two interacting subsystems. • Team uses evidence from mechanisms, structures, controls, materials, and performance testing. • Prototype is documented with sketches/CAD, test data, and revision history. • Final design review clearly defends the recommendation and next improvement.
Constraints	• Use approved classroom equipment, materials, and tool access. • Teams must manage roles, schedule, safety, and subsystem integration. • The final design must be demonstrable or clearly supported by prototype evidence.
Required Evidence	• Team concept and system architecture • Subsystem sketches/CAD/build evidence • Test plan and data • Revision log • Final presentation and engineering notebook package
Checkpoints	• Define the mission and system boundary • Create team concepts and combine ideas • Assign subsystems and build plan • Prototype and test • Integrate, revise, and present

Final Design Review Expectations

Your final review should clearly connect the problem, evidence, prototype decisions, testing results, limitations, and next recommended improvement. Strong teams show how data changed the design rather than only describing what they built.

Student Deliverable Reminder

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